



**LOMBARD ODIER**  
INVESTMENT MANAGERS

# Empirical Methods in Finance

Class 3 - Understanding the Linear Univariate Models (for good)

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Florian Ielpo, [f.ielpo@lombardodier.com](mailto:f.ielpo@lombardodier.com)

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Lombard Odier Investment Management

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# Introduction

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## Outline of Class 3

- Refreshing your knowledge of OLS estimates
- Briefly discuss the Fama French model
- Maximum Likelihood estimation again
- A practical example: the Taylor Rule

# Why is estimation so important in economics and finance?

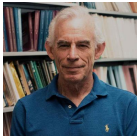
- Economists and financial engineers produce many models with sets of unknown parameters.
- These models need to be made as "realistic" as possible.
- We therefore need to squeeze them so that they can resemble reality.
- How do we do that?  $\Rightarrow$  we need to "calibrate" their parameters in a way that make them the closest they can get to reality.
- Many techniques: OLS, GMM, Maximum Likelihood, Simulated Method of Moments, Indirect Inference, Instrumental Variables... When use what?
- Many models: CAPM, Gordon and Shapiro, Taylor rule, Growth models, Vasicek models... they all incorporate some time series dimensions that needs to be dealt with.
- Another way of thinking of timeseries analysis: exploring linear relationships between economic and financial variables.

## Two ways to do time series analysis:

- Starting from a theoretical model, and trying to estimate its parameters given a time period.  
⇒ sometimes referred to as *calibration*.
- Starting from a time series, exploring its salient feature and creating a model around them.  
⇒ Usually called the "Sims Criticism", after its article "Macroeconomics and Reality." *Econometrica* 1980, 48: 1 (January): 1-48.

# Some of the Econometricians We Are Going To Talk About

Christophe Sims



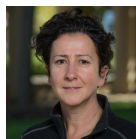
Clive Granger



Robert Engle



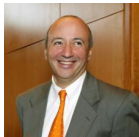
Monika Piazzesi



Andrew Patton



Yacine Aït Sahalia



Lars Hansen



Christian Gourieroux



Some guy I met on the campus



The guy I'm replacing



Darrell Duffie



Tim Bollerslev



# A first toy example

$$r_t = \alpha + \beta_1 x_t^{(1)} + \beta_2 x_t^{(2)} + \dots + \beta_p x_t^{(p)} + \sigma \epsilon_t \quad (1)$$

defines the linear model with:

- $\alpha$  the constant term, aka the intercept
- $r_t$  is known as the 'endogenous' variable: it depends on other factor
- $x_t^{(j)}$  are the 'exogenous' variables: they *explain*  $r_t$  without anything else explaining them
- $\epsilon_t$  is what the exogenous variables cannot explain: model error, observation errors...
- the  $\beta_j$  turn  $x_t^{(j)}$  into  $r_i$

$\Rightarrow x_t^{(j)}$  are "factors" in the financial literature.



# The Fama-French model

$$r_t = \alpha + \beta_1 f_t^{(Market)} + \beta_2 f_t^{(SMB)} + \beta_3 f_t^{(HML)} + \sigma \epsilon_t \quad (2)$$

This model is widely used to explain the performance of individual stocks across three market factors:

- $r_t$  is the return on a given individual stock.
- Market is the market factor (what globally happens across equity indices)
- SMB stands for Small Minus Big: the return of the largest companies vs. the return on the smallest (size factor)
- HML stands for High Minus Low: the returns of the most expensive vs. the cheapest companies (value factor, high book to market value vs. low book to market value)
- $\epsilon_t$  is now whatever the model does not explain and/or uncertainty sources arising from individual stocks (idiosyncratic risk).

⇒ We still need to estimate the model, i.e. assigning the most realistic values to its parameters (the  $\beta$  and  $\alpha$ ).

There exists broadly three estimation technics that you have heard of:

- Ordinary Least Squares: assigning to parameters values such that the errors between reality and model are as small as possible.
- Maximum Likelihood: assigning to parameters values such that the model has the greatest achievable probability of being right.
- Generalized Method of Moments: assigning to parameters values that make sure that the moments of the model are as close as possible to the moment of the phenomenon we are trying to describe.

In this class: mainly Maximum Likelihood as for the model class we will be dealing with, it is usually the strongest method.

## OLS refresher

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# The Least Square Estimation to the Linear Model

OLS (Ordinary Least Square) estimators are easily obtained when presented in the form of matrices. We can rewrite the linear model in a matrix form:

$$\begin{pmatrix} r_1 \\ r_2 \\ \vdots \\ r_n \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 2 \\ \vdots \\ 1 \end{pmatrix} + \beta_1 \begin{pmatrix} x_{1}^{Market} \\ x_{2}^{Market} \\ \vdots \\ x_n^{Market} \end{pmatrix} + \beta_2 \begin{pmatrix} x_{1}^{SMB} \\ x_{2}^{SMB} \\ \vdots \\ x_n^{SMB} \end{pmatrix} + \beta_3 \begin{pmatrix} x_{1}^{HML} \\ x_{2}^{HML} \\ \vdots \\ x_n^{HML} \end{pmatrix} + \sigma \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix} \quad (3)$$

$$\Leftrightarrow \begin{pmatrix} r_1 \\ r_2 \\ \vdots \\ r_n \end{pmatrix} = \begin{pmatrix} 1 & x_{1}^{Market} & x_{1}^{SMB} & x_{1}^{HML} \\ 1 & x_{2}^{Market} & x_{2}^{SMB} & x_{2}^{HML} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_n^{Market} & x_n^{SMB} & x_n^{HML} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} + \sigma \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix} \quad (4)$$

$$\Leftrightarrow R = X\beta^T + \sigma\epsilon \quad (5)$$

# The Least Square Estimation to the Linear Model

The OLS estimator solves the following least square criterion:

$$\beta_{OLS} = \min_{\beta} (R - X\beta^{\top})^{\top} (R - X\beta^{\top}) \quad (6)$$

Key matrix operations:

- let  $y = Ax$  be a matrix product,  $x$  being a  $p \times 1$  matrix of parameters and  $A$  a  $n \times p$  matrix. Then:  $\partial_x y = A$ .
- let  $y = x^{\top} Ax$  be a matrix product,  $x$  being a  $p \times 1$  matrix of parameters and  $A$  a  $n \times p$  matrix. Then:  $\partial_x y = x^{\top} (A^{\top} + A)$ . If  $A$  is symmetrical, then  $\partial_x y = 2x^{\top} A$ .

The least square estimate solves:

$$\min_{\beta} R^{\top} R - R^{\top} X \beta^{\top} - \beta X^{\top} R + \beta X^{\top} X \beta^{\top} \quad (7)$$

# The Least Square Estimation to the Linear Model

Opening brackets yields:

$$(R - X\beta^\top)^\top (R - X\beta^\top) = (R - \beta X^\top)(R - X\beta^\top) \quad (8)$$

$$= R^\top R - R^\top X\beta^\top - \beta X^\top R + \beta X^\top X\beta^\top \quad (9)$$

Differentiating with respect to  $\beta$  yields:

$$\partial_\beta R^\top R - R^\top X\beta^\top - \beta X^\top R + \beta X^\top X\beta^\top = -X^\top R - X^\top R + 2\beta X^\top X \quad (10)$$

Finally solving yields

$$\beta = (X^\top X)^{-1}(X^\top R) \quad (11)$$

# The Least Square Estimation to the Linear Model

The OLS estimator is a cocktail of random variable, which means it is also a random variable and its distribution is usually Gaussian either as:

- conditionally upon the knowledge of the  $x_t$ s, it is Gaussian because of the Central Limit Theorem (asymptotic normality).
- or because the conditional distribution of  $\epsilon$  is Gaussian (finite distance normality).

$$\hat{\beta}^\top = (X^\top X)^{-1}(X^\top R) \quad (12)$$

$$= (X^\top X)^{-1}(X^\top (X\beta^\top + \sigma\epsilon)) \quad (13)$$

$$= (X^\top X)^{-1}X^\top X\beta^\top + \sigma(X^\top X)^{-1}X^\top \epsilon \quad (14)$$

This implies:

$$E[\hat{\beta}^\top] = \beta^\top \quad V[\hat{\beta}^\top] = \sigma^2 V[(X^\top X)^{-1}X^\top \epsilon] = \sigma^2(X^\top X)^{-1} \quad (15)$$

# The Student Test

A test to check whether a parameter's value is equal to an assumed value.

$$H_0 : \beta_i = x \quad (16)$$

$$H_1 : \beta_i \neq x \quad (17)$$

Under  $H_0$ ,

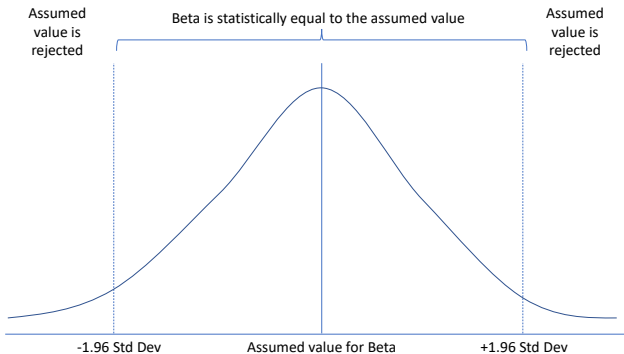
$$\beta_i \sim N(0, \sigma_{\beta_i}) \quad (18)$$

Therefore, 95% of the time, if  $\beta_i$  is really equal to  $x$  as assumed under  $H_0$ , the following statement should be true:

$$\frac{\hat{\beta}_i - x}{\sigma_{\beta_i}} \in [-1.96 + 1.96] \quad (19)$$



# The Student Test



# The Least Square Estimation to the Linear Model

## The R<sup>2</sup>

- R-squared is a statistical measure of how close the data are to the fitted regression line.
- It is also known as the coefficient of determination, or the coefficient of multiple determination for multiple regression.
- 0% indicates that the model explains none of the variability of the response data around its mean.

$$R^2 = 1 - \frac{(R - \hat{R})^\top (R - \hat{R})}{V(R)} \quad (20)$$

## Fama-French

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# Fama French Model Estimated

Fama French Model:

$$r_t = \alpha + \beta_1 f_t^{(Market)} + \beta_2 f_t^{(SMB)} + \beta_3 f_t^{(HML)} + \sigma \epsilon_t \quad (21)$$

Estimated using daily data (taken from Fama-French) over the 2011-2020 period.

Output Table:

|         | $\beta_3$ | $\beta_2$ | $\beta_1$ | $\alpha$ | R2   |
|---------|-----------|-----------|-----------|----------|------|
| Apple   | -0,55     | -0,30     | 1,13      | 0,00     | 0,48 |
| Netflix | -1,02     | 0,32      | 1,10      | 0,00     | 0,18 |
| Pfizer  | -0,11     | -0,30     | 0,76      | 0,00     | 0,42 |

Over that period:

- Apple and Netflix have been aggressive stocks, Pfizer more defensive.
- 18 to 48% of daily returns are explained by Fama-French, Netflix bearing the highest idiosyncratic risk.
- Pfizer is defensive, rather big and anti-value (more growth). Netflix has been on the Small side.

# Fama French Model Estimated

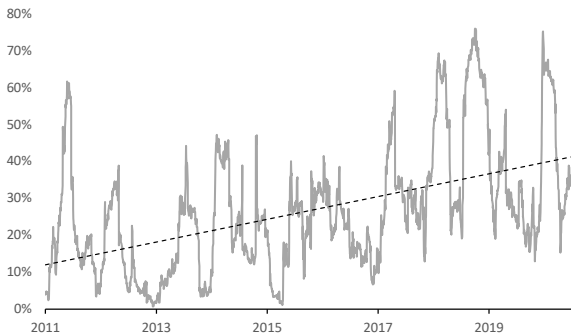
## Output of the Fama-French Regression with Netflix (2010-2020)

|                                             | $\beta_3$ | $\beta_2$ | $\beta_1$ | $\alpha$ |
|---------------------------------------------|-----------|-----------|-----------|----------|
| Estimate                                    | -1,02     | 0,32      | 1,10      | 0,00     |
| Standard Deviation                          | 0,09      | 0,11      | 0,05      | 0,00     |
| Estimate/Standard Deviation Ratio (Student) | -11,43    | 2,95      | 20,17     | 1,59     |
| R2                                          | 0,18      |           |           |          |

### Conclusions:

- Netflix has been over the period a growth small stock rather aggressive.
- All parameters are statistically significant but the  $\alpha$ .
- The 3-factor model explains 18% of Netflix's returns.

## Rolling estimation of Netflix's R2



## Maximum likelihood

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Probability of observing a draw  $x_1, x_2, \dots, x_n$  given a parametric model:

$$P(x_1, x_2, \dots, x_n | \theta) = P(\theta | x_1, x_2, \dots, x_n) \quad (22)$$

ML method means picking  $\theta$  such that this probability is max, i.e. so that it can be as likely as possible that this sample has been drawn from this parametric model.



Problem: for most model impossible to compute this probability given that:

- for a continuous distribution this probability is equal to 0.
- for most models, the joint distribution cannot be derived (need to find a trick there).

⇒ yet ML makes best use of all the information about the distribution.

Solution: probability is proportional to density

$$P(x_1|\theta) = \int_{x_1-\epsilon}^{x_1+\epsilon} f(x)dx \quad (23)$$

Instead of joint probability, use joint density:

$$f(x_1, x_2, \dots, x_n|\theta) \quad (24)$$

With iid data:

$$f(x_1, x_2, \dots, x_n|\theta) = \prod f(x_t|\theta) \quad (25)$$

Maximizing a product is numerically complex, better to maximize the log of it

$$\max \log \prod f(x_t|\theta) \quad (26)$$

Which then becomes:

$$\max \sum \log f(x_t|\theta) \quad (27)$$

For most distribution, density uses exponential  $\Rightarrow$  tractable expression.

ML estimates solve for i.i.d. observations:

$$\max_{\theta} \sum_{i=1}^n \log f(x_i|\theta) \quad (28)$$

or equivalently:

$$\sum_{i=1}^n \partial_{\theta} \log f(x_i|\theta) = 0 \quad (29)$$

Several important hypotheses:

- Identification:  $\forall \theta \neq \theta^*, \sum_{i=1}^n \log f(x_i|\theta) \neq \sum_{i=1}^n \log f(x_i|\theta^*)$ .  
One set of parameter, one likelihood value. Can be violated for several finance models such as Vasicek model, or in probit models...
- first three derivatives of  $\log f(x_t|\theta)$  are continuous and finite  $\forall \theta$
- $\mathbb{E}[\partial_\theta \log f(x_t|\theta)] < \infty$  and  $\mathbb{E}[\partial_{\theta^2} \log f(x_t|\theta)] < \infty$
- $|\partial_{\theta^3} \log f(x_t|\theta)| < h$ , with  $\mathbb{E}[h] < \infty$ .

Last three conditions mean we deal with regular densities.

# Maximum likelihood

Why love maximum likelihood? Because of the asymptotic efficiency of its estimates.

ML estimates are:

- Consistency :  $\mathbb{E}[\hat{\theta}_{ML}] = \theta_0$  or  $\text{plim} \hat{\theta} = \theta_0$
- Asymptotic normality:  $\hat{\theta} \rightarrow N\left(\theta_0, [-\mathbb{E}[\partial_{\theta^2} \log L]]^{-1}\right)$
- Asymptotic efficiency:  $\hat{\theta}$  reaches the FDCR lower bound.
- Invariance: the ML estimate of  $\gamma_0 = g(\theta_0)$  is  $\hat{\gamma}_0 = g(\hat{\theta}_0)$

Third item means that when the ML conditions are granted, then ML estimates are the most efficient estimates of the world.

$I_{\theta} = \mathbb{E}[\partial_{\theta^2} \log L]$  is called the information matrix, as it informs its user on the variance of the estimates around the optimal value.

Also:  $-\mathbb{E}[\partial_{\theta^2} \log L] = \mathbb{E}[\partial_{\theta} \log L \times \partial_{\theta} \log L]$  for a single parameter – or its matrix equivalent.

# Pseudo/Quasi Maximum likelihood

What happens if the model is misspecified?

⇒ pseudo maximum likelihood: under some circumstances, the ML estimates of wrongly specified model remains consistent! Globally,  $f(\cdot)$  must belong to the exponential family.

$$\sqrt{n}(\hat{\theta}_{PML} - \theta_0) \rightarrow N(0, H^{-1}\Phi H^{-1}) \quad (30)$$

with

$$\Phi = \text{Cov} \left( \frac{\partial \log f(x_t)}{\partial \theta} \right) \quad (31)$$

Usually:  $\Phi$  estimated via  $\frac{1}{n} \sum_{i=1}^n \left[ \frac{\partial \log f(x_t|\theta_0)}{\partial \theta_0} \right] \left[ \frac{\partial \log f(x_t|\theta_0)}{\partial \theta_0} \right]^T$

When  $\Phi = -H(\theta_0)$  then PML and ML estimates are the same.

# Conditional Densities

In the case of timeseries, serial dependencies are common things:  $f(x_t)$  are *not* i.i.d.

Solution: conditioning on past information.

$$f(x_1, x_2, \dots, x_n) = f(x_1) \times f(x_2|x_1) \times \dots f(x_n|x_1, \dots, x_{n-1}) \quad (32)$$

The loglikelihood is then the sum of the conditional log-likelihoods:

$$\text{Log}L = \sum_{t=1}^n \log f(x_t|x_1, \dots, x_{t-1}) \quad (33)$$

Usually denoted:

$$\text{Log}L = \sum_{t=1}^n \log f(x_t|\underline{x_{t-1}}) \quad (34)$$



# The Change in Variable Theorem

The Change in Variable Theorem is essential to compute the conditional distribution for a lot of models.

**Theorem** Suppose  $X$  is continuous with probability density function  $f_X(x)$ . Let  $y = h(x)$  with  $h$  a strictly increasing continuously differentiable function with inverse  $x = g(y)$ .

Then  $Y = h(X)$  is continuous with probability density function  $f_Y(y)$  given by

$$f_Y(y) = f_X(g(y))g'(y) \quad (35)$$

# The Change in Variable Theorem

Example: Let  $\epsilon_t \sim N(0, 1)$ . Find the distribution of  $Y_t = a + \sigma\epsilon_t$ .

Solution:

$$f_Y(y) = f_\epsilon\left(\frac{y-a}{\sigma}\right) \times \left(\frac{y-a}{\sigma}\right)' \quad (36)$$

Therefore

$$f_Y(y) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2}\left(\frac{y-a}{\sigma}\right)^2\right) \quad (37)$$

this implies  $Y \sim N(a, \sigma)$

# Maximum Likelihood Estimation

Considering the following model:

$$r_i = \alpha + \beta x_t + \sigma \epsilon_i \quad (38)$$

with  $\epsilon_i$  following a Gaussian distribution  $N(0, 1)$ .

Step 1: using the change of variable theorem, find the distribution of  $r_i$ .

$$f_{r_i}(r_i) = f_{\epsilon_i} \left( \frac{r_i - \mu}{\sigma} \right) \times \frac{\partial \frac{r_i - \mu}{\sigma}}{\partial r_i} \quad (39)$$

From that result,  $r_i \sim N(\alpha + \beta x_t, \sigma)$ .

Step 1: write the log-likelihood of the model and derive the FOC:

$$\log L(r_1, r_2, \dots, r_n | \mu, \sigma) = -\frac{n}{2} \log(2\pi) - n \log \sigma - \sum_{i=1}^n \frac{1}{2} \left( \frac{r_i - \alpha - \beta x_t}{\sigma} \right)^2 \quad (40)$$

# Maximum Likelihood Estimation

FOC:

$$\frac{\partial \log L}{\partial \beta} = \sum_{i=1}^n x_t \left( \frac{r_i - \alpha - \beta x_t}{\sigma} \right) = 0 \quad (41)$$

$$\frac{\partial \log L}{\partial \alpha} = \sum_{i=1}^n \left( \frac{r_i - \alpha - \beta x_t}{\sigma} \right) = 0 \quad (42)$$

$$\frac{\partial \log L}{\partial \sigma} = -\frac{n}{\sigma} + \sum_{i=1}^n \frac{(r_i - \alpha - \beta x_t)^2}{\sigma^3} = 0 \quad (43)$$

Which yields the following estimates:

$$\hat{\alpha} = \frac{1}{n} \sum_{i=1}^n (r_i - \beta_i x_t), \quad \hat{\beta} = \frac{\sum_{i=1}^n (r_i - \hat{\alpha} - \beta_i x_t)}{\sum_{i=1}^n x_t^2} \quad (44)$$

$$\hat{\sigma} = \frac{1}{n} \sum_{i=1}^n (r_i - \hat{\alpha} - \hat{\beta}_i x_t)^2 \quad (45)$$

## Model assessment and forecasting

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# How to assess the quality of a model?

## Two different measures:

1. In sample: the  $R^2$  ("R-square") looks into the quality of the fit:

$$R^2 = 1 - \frac{\mathbb{V}[\hat{\epsilon}_i]}{\mathbb{V}[r_i]} \quad (46)$$

Ranges from 0 to 1, close to one when the residuals have almost no variability left.

2. Out of sample:

- Root Mean Squared Errors =  $\sqrt{\frac{1}{n} \sum_i (r_i - \hat{r}_i)^2}$ , with  $\hat{r}_i$  the forecast variables  $r_i$ , with  $n$  forecasts.
- Mean Absolute Errors =  $\frac{1}{n} \sum_i |r_i - \hat{r}_i|$ .

## Some others tests people could ask you about

1. Fisher test: a significant test for all parameters in the meantime.  $H_0$  is all parameters are equal to 0. Under  $H_0$  the following test statistic follows a Fisher distribution:

$$F_{\text{test}} = \frac{R^2}{1 - R^2} \frac{n - p - 1}{p} \sim F(p, n - p - 1), \quad (47)$$

with  $n$  the number of observations and  $p$  the number of parameters.

2. Student test: testing for only one parameter's significance. Significance means "can this parameter be set to 0?".  $H_0 : \theta = 0$ , with  $\theta$  a given parameter. Under  $H_0$ ,

$$\frac{\hat{\theta}}{\sigma(\theta)} \sim N(0, 1), \quad (48)$$

provided that you have enough information.

3. Durbin and Watson test: an unusual test to check that errors are not *autocorrelated*. The test statistics:

$$d = \frac{\sum_i (\hat{\epsilon}_i - \hat{\epsilon}_{i-1})^2}{\sum_i \hat{\epsilon}_i^2}. \quad (49)$$

$d$  oscillates between 4 and 0. 2 means no autocorrelation.

Once the relationship is estimated, we can make forecasts:  $r_j = \beta x_j + \sigma \epsilon_j$  (simplified version with no intercept).

The 'forecast' error:  $e_j = y_j - \hat{y}_j = x_j(\beta - \hat{\beta}) + \sigma \epsilon_j$ , with the variance:

$$V(e_j) = x_j^2 V(\hat{\beta}) + \sigma^2 = x_j^2 \frac{\sigma^2}{\sum_{i=1}^n x_t^2} + \sigma^2 \quad (50)$$

Forecast errors are function of

- the estimation errors' variance.
- the forecast  $x_j$ 's values
- the volatility of the  $x_t$  as

$$V(e_j) = \sigma^2 \left( 1 + \frac{x_j^2}{\sum_{i=1}^n x_t^2} \right) \quad (51)$$



## Taylor rule application

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## A toy example: the Taylor rule

Hypothesized relationship between the Fed's decision rate and economic variables:

$$R_t = \alpha + \beta_\pi \pi_t + \beta_g g_t + \epsilon_t \quad (52)$$

with

1.  $\pi_t$  the inflation rate at time  $t$
2.  $g_t$  the growth rate of GDP at time  $t$
3.  $R_t$  the Fed's decision rate.

The model assumes that  $\epsilon_t$  is iid and that  $\text{Cov}(\epsilon_t, \epsilon_{t-1}) = 0$ : monetary policy shocks are not persistent.

The Taylor rule is a **policy reaction function**: it summarises how a central bank adjusts its policy rate when macro conditions change.

- **Stabilisation logic**: raise rates when inflation is above target and/or activity is above potential.
- **A benchmark, not a mechanical recipe**: used to discuss whether policy is tight or loose relative to a simple rule.
- **Operational link**: connects a theoretical model (inflation and output gap dynamics) to an observed instrument (the policy rate).

# Canonical Taylor (1993) rule

A common benchmark form is:

$$i_t = r^* + \pi_t + \phi_\pi (\pi_t - \pi^*) + \phi_y y_t^{gap},$$

where  $i_t$  is the nominal policy rate,  $r^*$  the equilibrium real rate,  $\pi^*$  the inflation target, and  $y_t^{gap}$  the output gap.

- In the original calibration,  $\phi_\pi \approx 0.5$  and  $\phi_y \approx 0.5$ .
- **Taylor principle:** for macro stability, the rule should satisfy  $\frac{\partial i_t}{\partial \pi_t} > 1$  (raise the nominal rate more than one-for-one with inflation).

# Theory origins: why rules, and why this structure?

The Taylor rule is a reduced-form summary of ideas that predate Taylor (1993):

- **Rules vs discretion** (time inconsistency): the case for systematic policy rather than purely discretionary actions.
- **Wicksellian logic**: stabilise the economy by adjusting the policy rate relative to a natural rate.
- **New Keynesian foundation**: with an IS curve and a Phillips curve, a sufficiently strong reaction to inflation helps deliver determinacy and anchors expectations.

Interpretation: it is a compact way to map the central bank's **loss function** (inflation stabilisation and real activity stabilisation) into a simple implementable rule.

# Academic debate: what is controversial in practice?

Even if the rule is conceptually clean, implementation raises hard questions:

- **Measurement:** output gaps and  $r^*$  are unobserved and revised heavily over time.
- **Forward-looking behaviour:** policy reacts to forecasts, not only to realised inflation and growth.
- **Policy inertia:** many central banks smooth rates, often modelled with a lagged policy rate.
- **Regime shifts:** reaction coefficients can change across chairs, mandates, and macro regimes.
- **Constraints:** effective lower bound, QE, and macro-prudential tools break the simple one-instrument framework.

# Empirical applications: why estimate a Taylor rule?

A fitted Taylor rule is used in many applied settings:

- **Describe stance:** compare the actual policy rate to a rule-implied benchmark.
- **Cross-country comparisons:** estimate reaction functions for the Fed, ECB, BoE, BoJ, etc.
- **Counterfactuals:** what would rates have been under alternative coefficients or targets?
- **Shock identification:** reaction-function residuals are often used as monetary policy shocks.
- **Forecasting:** simple rules are competitive baselines for short-horizon rate projections.

In practice, applied work often augments the toy rule with inertia and additional controls:

$$i_t = \alpha + \rho i_{t-1} + \beta_\pi \pi_t + \beta_y y_t^{gap} + \delta' z_t + u_t,$$

where  $z_t$  can include financial conditions, exchange rates, or forecast-based measures.

Key econometric point: once  $i_{t-1}$  is on the right-hand side, we must be careful about **serial correlation** in  $u_t$ .



## Serial correlation is not only about standard errors

A common reflex is to apply HAC (Newey-West) standard errors when residuals are autocorrelated.

- HAC fixes **inference** (standard errors) under weak conditions.
- But if autocorrelation comes from a **misspecified error process** that makes regressors endogenous (e.g., lagged  $i_{t-1}$ ), OLS can be **biased and inconsistent**.

We will now show precisely how an **omitted moving-average component** can create correlation between  $i_{t-1}$  and the error term, which mechanically shifts OLS estimates.

# General form of the asymptotic OLS bias

Consider a linear regression:

$$y_t = x_t' \beta + u_t.$$

The OLS estimator satisfies:

$$\hat{\beta} = \beta + (X'X)^{-1} X'u.$$

Under standard regularity conditions,

$$\text{plim } \hat{\beta} = \beta + Q^{-1}q, \quad Q = \mathbb{E}[x_t x_t'], \quad q = \mathbb{E}[x_t u_t].$$

So the OLS probability limit equals the truth only if  $\mathbb{E}[x_t u_t] = 0$ .

## Omitted MA(1) component: where endogeneity appears

Focus on the inertial part of a Taylor rule and assume the true data-generating process is:

$$i_t = \alpha + \rho i_{t-1} + u_t, \quad u_t = \varepsilon_t + \theta \varepsilon_{t-1}, \quad \varepsilon_t \stackrel{iid}{\sim} (0, \sigma_\varepsilon^2).$$

If we estimate the regression by OLS while **treating  $u_t$  as iid**, the regressor  $i_{t-1}$  is not exogenous:

$$\text{Cov}(i_{t-1}, u_t) = \text{Cov}(\alpha + \rho i_{t-2} + u_{t-1}, u_t) = \text{Cov}(u_{t-1}, u_t) = \theta \sigma_\varepsilon^2 \neq 0.$$

This single covariance term is enough to generate **biased OLS coefficients**.

## Deriving the OLS bias precisely (ARMA(1,1) example)

OLS for  $\rho$  in  $i_t = \alpha + \rho i_{t-1} + u_t$  can be written:

$$\hat{\rho} = \frac{\sum_{t=1}^T (i_{t-1} - \bar{i})(i_t - \bar{i})}{\sum_{t=1}^T (i_{t-1} - \bar{i})^2} = \rho + \frac{\sum_{t=1}^T (i_{t-1} - \bar{i})u_t}{\sum_{t=1}^T (i_{t-1} - \bar{i})^2}.$$

Under stationarity, as  $T \rightarrow \infty$ ,

$$\text{plim } \hat{\rho} = \rho + \frac{\mathbb{E}[i_{t-1}u_t]}{\mathbb{E}[i_{t-1}^2]} = \rho + \frac{\theta\sigma_\varepsilon^2}{\text{Var}(i_t)}.$$

For an ARMA(1,1) process  $i_t = \rho i_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}$  with  $|\rho| < 1$ ,

$$\text{Var}(i_t) = \frac{\sigma_\varepsilon^2 (1 + \theta^2 + 2\rho\theta)}{1 - \rho^2}.$$

Therefore,

$$\text{plim } \hat{\rho} = \rho + \theta \frac{1 - \rho^2}{1 + \theta^2 + 2\rho\theta}$$

The bias is zero only if  $\theta = 0$ . Its sign follows the sign of  $\theta$ .

## Practical takeaway for Taylor-rule estimation

If your residuals suggest MA-type dynamics, it is not enough to use HAC standard errors.

- **Correct the model:** estimate with ARMA errors (GLS / ML) so that the residual process is explicitly modelled.
- **Or change the estimator:** use IV/GMM when regressors become endogenous due to dynamics.
- **Interpretation:** autocorrelation can be policy inertia, omitted variables, or measurement error - the econometric fix should match the economic story.

## Practical application

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# What this notebook does

The notebook is split into three parts, aligned with EMiF-Class3:

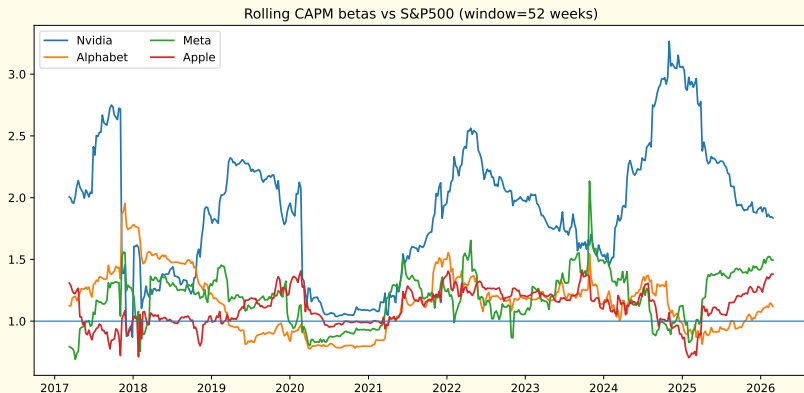
1. **US equities** – CAPM and multi-factor betas, rolling stability
2. **Swiss equities** – CAPM betas (SMI as market), rolling stability
3. **Taylor rule** – OLS first, then MA(1) errors by maximum likelihood; bias when MA is ignored

## US equities – CAPM betas (excess returns)

|          | alpha | beta_mkt | R2    |
|----------|-------|----------|-------|
| asset    |       |          |       |
| Nvidia   | 0.007 | 1.688    | 0.409 |
| Meta     | 0.001 | 1.135    | 0.293 |
| Apple    | 0.002 | 1.111    | 0.467 |
| Alphabet | 0.002 | 1.047    | 0.430 |



# US equities – rolling CAPM betas



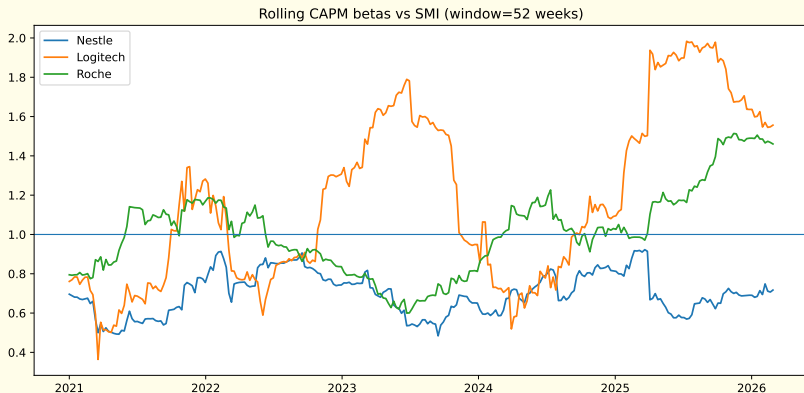
# US equities – multi-factor betas (Market, Value, Growth, Quality)

|          | alpha | beta_S&P500 | beta_G | beta_V | beta_Q | R2    |
|----------|-------|-------------|--------|--------|--------|-------|
| asset    |       |             |        |        |        |       |
| Nvidia   | 0.005 | 4.256       | -0.151 | -2.851 | 0.252  | 0.543 |
| Alphabet | 0.002 | -2.840      | 3.136  | 1.241  | -0.656 | 0.548 |
| Meta     | 0.001 | -3.741      | 3.776  | 1.442  | -0.530 | 0.393 |
| Apple    | 0.001 | 4.093       | -1.194 | -2.138 | 0.295  | 0.530 |

## Swiss equities – CAPM betas (SMI as market)

|          | alpha  | beta_mkt | R2    |
|----------|--------|----------|-------|
| asset    |        |          |       |
| Logitech | 0.000  | 1.068    | 0.211 |
| Roche    | -0.000 | 0.960    | 0.455 |
| Nestle   | -0.001 | 0.726    | 0.403 |

# Swiss equities – rolling CAPM betas



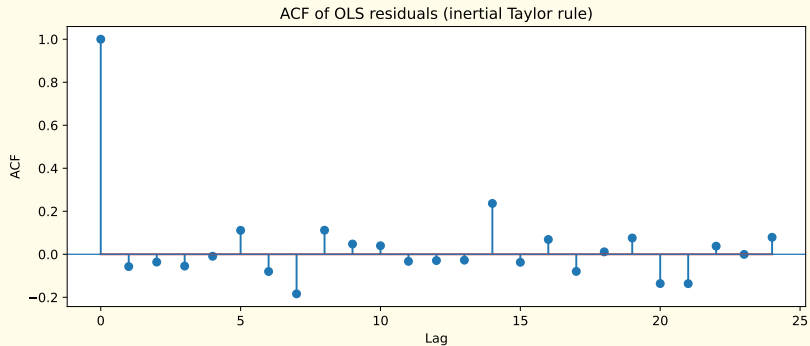
## Taylor rule – OLS (inertial specification)

Inertial Taylor rule:

$$i_t = \alpha + \phi_\pi \pi_t + \phi_y y_t + \rho i_{t-1} + u_t$$

|           | beta  | se    | t       |
|-----------|-------|-------|---------|
| const     | 0.065 | 0.047 | 1.400   |
| Inflation | 0.040 | 0.012 | 3.292   |
| OutputGap | 0.045 | 0.013 | 3.519   |
| i(t-1)    | 0.959 | 0.009 | 101.155 |

# Residual autocorrelation (OLS)



# MA(1) likelihood and numerical estimation

We assume:

$$u_t = \varepsilon_t + \theta \varepsilon_{t-1}, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2)$$

Conditional recursion:

$$\varepsilon_t = u_t - \theta \varepsilon_{t-1}$$

We estimate  $(\beta, \theta, \sigma)$  by minimizing the negative log-likelihood (simple recursion + `scipy.optimize.minimize`).

**Result (OLS vs MLE(MA1)):**

|           | OLS   | MLE(MA1) | diff (MLE-OLS) |
|-----------|-------|----------|----------------|
| const     | 0.065 | 0.059    | -0.006         |
| Inflation | 0.040 | 0.037    | -0.003         |
| OutputGap | 0.045 | 0.044    | -0.001         |
| i(t-1)    | 0.959 | 0.963    | 0.004          |

## Bias illustration – Monte Carlo (ignore MA vs account for MA)

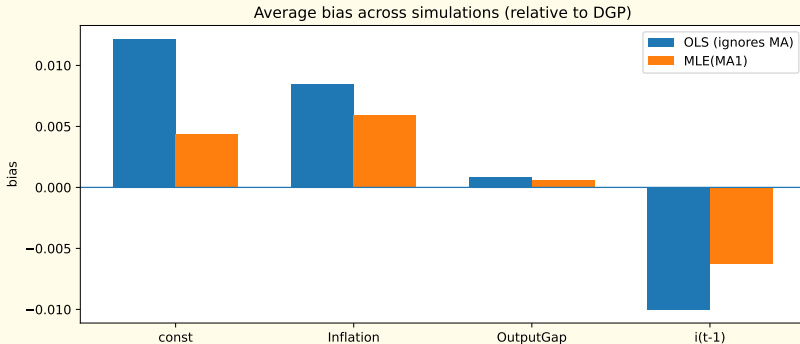
We simulate the inertial Taylor rule recursively (so  $i_{t-1}$  depends on past shocks) with MA(1) errors. Then we estimate:

- OLS (misspecified: ignores MA)
- MLE(MA1) (correct)

|           | true (DGP) | mean OLS | bias OLS | mean MLE | bias MLE |
|-----------|------------|----------|----------|----------|----------|
| const     | 0.059      | 0.071    | 0.012    | 0.063    | 0.004    |
| Inflation | 0.037      | 0.045    | 0.008    | 0.043    | 0.006    |
| OutputGap | 0.044      | 0.045    | 0.001    | 0.045    | 0.001    |
| $i(t-1)$  | 0.963      | 0.953    | -0.010   | 0.957    | -0.006   |



# Bias illustration – summary chart



# Main takeaways

1. **Betas are not constants** – rolling estimates show stability (or instability) over time.
2. **Taylor rule residuals can be autocorrelated** – check it (ACF is a quick diagnostic).
3. **Ignoring MA structure can move coefficients** in dynamic regressions (with inertia), because lagged rates embed past shocks.
4. Coding a **simple likelihood** (few lines) is enough to estimate an MA(1) error model and compare to OLS.